

7. AI for Research

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The Research Loop, Assisted

Find

- Discover and track the literature
- Follow citations forward and back

Organize

- A private, searchable library of what you have read
- Search by keyword *and* by meaning

Write & review

- Draft and edit with field-aware guidance
- Referee your own work before a journal does

Four moves, each with a tool you can install and use today. AI does the search, the recall, and the first pass; the judgment stays yours.

Literature Search

Finding and Tracking Work

Ask, don't keyword

- Describe what you want in plain language
- Discovery across **OpenAlex** and **Semantic Scholar** — millions of papers, open metadata
- Get papers and abstracts, not just a list of links

Follow the citations

- What does this paper cite? Who cites it?
- Trace a result forward to its critics and back to its roots
- Surface the branch of the literature you were missing

The web search from Day 2, pointed at the scholarly record. It finds candidates; you still decide what is relevant.

Knowledge Databases

Your Own Library, Searchable

What it is

- A local, private store of your PDFs and notes
- Full-text keyword *and* semantic (meaning) search
- Import a whole Zotero library, or a folder of PDFs

Why it helps

- “What do I have on X?” — answered in seconds
- Notes tied to the papers that prompted them
- Export BibTeX straight into your paper

The `litdb` skill backs this — your corpus first, external sources second. It runs locally, so your library and notes never leave your machine.

Writing

Drafting and Editing

Field-aware help

- Structure, argument, and clarity for an economics paper
- Synthesized from 50+ guides by working economists
- Section by section: intro, results, response to referees

Sound human, not machine

- Vary the rhythm; drop the AI tells
- Let the argument drive the structure, not a template
- Edit an existing draft, or draft from your notes

The econ-write and human-write skills. They do not invent your findings — they help you say what you found, well.

Reviews

Referee Before the Referee

What it produces

- A rigorous, source-grounded referee report on a draft
- Separate editing notes and a prioritized revision plan
- The objections a real reviewer will raise — first

Others have built these too

- Pedro Sant'Anna: a full paper-lifecycle suite, adversarial critic-fixer loop
- Lee Crawford: tightly-scoped review + replication-audit skills
- Installable, mostly economists — Claude as **copilot, not pilot**

The econ-review skill does this for your own drafts. You own the question and the identification; the skill enforces the discipline.

The Risk That Matters Most

The biggest academic danger with AI is **fabricated references** and claims a cited source does not actually support. The better research skills attack this head-on.

Verify every citation

Checked against
Semantic Scholar,
OpenAlex, Crossref,
arXiv — bogus DOIs
caught before export

Audit every claim

Does the source
actually support the
sentence? High-
severity violations
block the output

Flag contamination

Detect citations that
may themselves be AI-
invented

For empirical or b-school research, this is the line between a demo and something you can responsibly put your name on.

Does It Actually Work?

The benchmark

- 54 social-science papers, tested for whether a coding agent could reproduce the published findings from the paper alone
- Claude Code: **~93%** at the individual-step level
- But only **~78%** at the whole-paper level

What the gap teaches

- It nails each analysis step
- One broken step anywhere still fails the full reproduction
- Exactly where human oversight stays essential

The 93/78 split is the honest picture: extraordinary at the parts, not yet trustworthy end to end without a human checking the chain.

The Payoff

AI handles the **search, the recall, and the first pass** — finding the literature, remembering what you read, drafting the prose, refereeing the draft. You decide **what matters and whether it is right.**

Breakout

Questions to Discuss

- Where in your own research does recall of the literature slow you down most?
- What would you keep in a private knowledge base that you would never put in a public tool?
- Which is riskier to hand to AI — the drafting, or the reviewing? Why?
- Given the 93/78 result, which parts of your empirical work would you let AI run, and which would you always check yourself?